MILITARY SPECIFICATION

TEST SET, RADIO AN/ARM-28

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide facilities for bench testing of the proportional control radio receiver amplifiers of the R-602/ARW type and proportional control equipment of the KY-133/ARW-66 type.

1.2 Classification - The Test Set, Radio AN/ARM-28 covered by this specification shall be of one type and shall consist of the following:

Units	Type Designation	Max. Allow Size (Inches) W	H	L	Max. Allow. Wt. Lbs.	Appl. Par.
Test Set, Radio	TS-(*)/ARM-28	17-1/2	13-5/8	26-5/8	40	3 3.1
Cover, Test Set	CW-(*)/ARM-28	17-1/2	4	26-5/8	8	3 3.2
Cable Assembly, Power, Electrical	CX-(*)/U(8'-0")			96		3.3.3
Cable Assembly, Special Purpose, Electrical	CX-(*)/U(5'-0")			60		3.3.4
Cable Assembly, Special Purpose, Electrical	CX-(*)/U(5'-0")			60		3 3.5
Mount, Servo Test	MT-(*)/U	3-3/4	3-1/4	6-3/4		3 3.6
Mount, Gyro Test	MT-(*)/U	3-3/4	3-1/4	6-3/4		3 3.7

NOTE: The parentheses (*), in the type designation throughout this specification, shall be replaced by either a number or letter furnished by the procuring agency upon application by the contractor in accordance with the requirements of paragraph 3.2.10 after each contract has been awarded. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:26Z
Check the source to verify that this is the current version before use.

1.3 Accessories - The equipment shall operate with the following accessories, which are not to be supplied as part of this equipment .

Power Supply	28 V DC, 10 AMP
Control Box C-1395/ARW-66	
Control Box C-1396/ARW-66	
Signal Generator AN/ARM-19	

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-152	Treatment, Moisture and Fungus Resistant, of Communications, Electronic and Associated Electrical Equipment, General Process for
MIL-M-6	Meters, Electrical Indicating Panel Type
MIL-T-945	Test Equipment, For Use with Electronic Equipment, General Specification
MIL-P-642	Plugs, Telephone
MIL-C-3806	Connectors "BNC" For Radio Frequency Cables
MIL-E-4682	Electron Tubes, Choice and Application of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-C-5015	Connectors; Electrical "AN" Type
MIL-I-6181	Interference Limits and Tests; Aircraft Electrical and Electronic Equipment
MIL-T-5422	Testing; Environmental, Aircraft Electronic Equipment
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment
MIL-T-18303	Test Procedures, Preproduction and Inspection; Format for
MIL-P-17555	Preparation for Delivery of Electronic Equipment: Accessories, Auxiliary Equipment, and Associated Repair (Maintenance) Parts

Bureau of Aeronautics

SAR-300	Engineering Change Proposals, Electronic Material Changes and Electronic Material Bulletins; Preparation of

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:26Z
Check the source to verify that this is the current version before use.

STANDARDS

MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-129	Marking for Shipment and Storage

2.2 Availability of Documents:

(1) When requesting specifications, publications and drawings refer to both title and number. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable specifications and publications may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Philadelphia 11, Pennsylvania. Attention: Code ODPT

(3) Copies of applicable Bureau of Aeronautics drawings may be obtained upon application to Bureau of Aeronautics, Navy Department, Washington 25, D. C., Attention: Avionics Division.

2.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring agency.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern.

(1) Parts and materials as approved by specification MIL-E-5400 shall be given first consideration.

(2) When the contractors can demonstrate that the use of standard AN or MIL parts or materials will not fulfill the design objectives because of size, weight, performance or other reasons, materials and parts shall be used which most nearly meet or exceed the requirements of the respective standard specification.

(3) When a non-standard part is used, the meeting of general material requirements such as non-inflammability, fungus resistance, non-toxicity, etc., and, the meeting of environmental conditions as required by specification MIL-T-945 shall be of prime consideration

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:26Z
Check the source to verify that this is the current version before use.

(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass material, electrical and environmental tests. Any assembly built to be expended shall be approved first by the procuring agency.

3.1.1 Non-Standard Part Approval and Material Approval - Approval for the use of non-standard parts and materials shall be obtained as outlined in specification MIL-E-5400 except as follows:

(1) Mechanical parts are expected to meet the material requirements of MIL-E-5400, and shall be submitted as non-standard parts only when the material requirements are not met.

(2) Non-standard parts, as defined in specification MIL-E-5400 of an electrical nature must be approved before the preproduction model of the equipment is delivered to the Bureau of Aeronautics. The request for approval must be forwarded by the contractor before the part is purchased or fabricated. In addition to or in lieu of the NON-Standard Part Data Sheet, the contractor is requested to submit itemized lists of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.

3.1.2 Choice of Tube Types - Electronic tubes used in the equipment shall be chosen in accordance with the following requirements and in the sequence noted:

(1) MIL-E-4682

(2) The use of any tube not on these acceptable listings must be approved by the Bureau of Aeronautics, however, before consideration for approval, the contractor must submit the following information:

(a) Quantitative data on the electrical advantages and mechanical adequacy of the proposed tube over the most nearly equivalent type on the acceptable listings.

(b) Test data and performance limits of the tube which must be maintained to insure satisfactory and reliable performance in the equipment.

(c) Information as to whether the tube manufacturer has received Military or Navy qualification approval for the particular tube.

3.1.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives to be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the Bureau of Aeronautics for consideration. Each request should be accompanied by complete supporting information.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:26Z
Check the source to verify that this is the current version before use.

3.2 General Requirements

3.2.1 General - This equipment shall fulfill all applicable design and performance requirements of specification MIL-T-945, unless otherwise specified herein.

3.2.2 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions or reasonable combinations of service conditions as specified in MIL-T-945, unless otherwise specified herein.

3.2.2.1 Shock - Equipment shall be capable of meeting shock test specified in specification MIL-E-5400.

3.2.3 Total Weight - The total weight of the equipment including cables, and transit case, shall be a minimum consistent with good design and shall not exceed 40 pounds.

3.2.4 Operational Stability - The equipment shall operate with optimum performance for 200 hours, continuously or intermittently, without necessity for readjustment of any controls which are non-accessible during flight.

3.2.5 Operating Life

3.2.5.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 200 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.5.2 Total Operating Life - The equipment shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.6 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with specification MIL-T-945.

3.2.7 Input Power Requirements - The equipment shall operate in a satisfactory manner from a DC power supply having regulation characteristics such that the output voltage shall be not less than 30 volts -0.5 volts when a 10 amp load is applied. Rectifier power supplies not meeting these requirements shall have a suitable battery connected across the 30 V terminals.

3.2.8 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices shall be held to a minimum.

3.2.8.1 Under Voltage Protection - The equipment shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within limits.

3.2.9 Moisture and Fungus Resistant Treatment - Equipments shall be fungus proofed by selection of parts and materials as non-nutrient for fungus or the parts and materials shall be treated prior to their use in the equipment so that overall spraying of the equipment is not necessary. Overall spraying of the equipment in accordance with MIL-T-152 shall be made in the event selection of parts and materials described above is not possible.

3.2.10 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with the requirements of specification MIL-N-18307.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:26Z
Check the source to verify that this is the current version before use.

3.2.11 Interchangeability - The interchangeability requirements between like assemblies, subassemblies and replaceable parts as specified in MIL-T-945 shall be extended to require interchangeability of units of this equipment with the latest model of any previously produced equipment. Such interchangeability shall be measured against a model, manufacturing drawings, or other technical information as provided by the Bureau of Aeronautics for the purpose. In the event that requirements of MIL-T-945 conflict with the requirements for interchangeability, the interchangeability requirements shall govern.

3.2.12 Radio Interference Control - The generation of radio interference by the equipment and vulnerability of the equipment to radio interference shall be controlled within the limits of specification MIL-I-6181.

3.2.13 Workmanship - Workmanship, fabrication and manufacturing processes shall conform to the requirements of MIL-T-945.

3.3 Detail Requirements

3.3.1 Test Set, Radio TS-(*)/ARM-28 - The Radio Test Set TS-(*)/ARM-28 shall meet the following requirements:

3.3.1.1 Function - The Radio Test Set TS-(*)/ARM-28 with accessories noted in paragraph 1 3 shall provide means of testing KY-133/ARW-66 coder equipments complete with Control Box C-1395/ARW-66 or Control Box C-1396/ARW-66 and R-602/ARW Receiver Amplifiers, for conformance to the requirements of their respective specifications.

3.3.1.2 Form Factor and Mounting - The Test Set shall be housed in a combination type case constructed of aluminum in accordance with specification MIL-T-945 and shall be watertight in accordance with the requirements of MIL-STD-108. The cover shall be detachable and shall be secured by trunk type latches. The dimensions of the test set without cover, shall not exceed 17-1/2 inches wide, 13-5/8 inches high, and 26-5/8 inches long.

3.3.1.3 Weight - The test set with cables shall not exceed 40 pounds in weight.

3.3.1.4 Contents of Unit - The following items and circuits shall be provided in the Test Set:

3.3.1.4.1 Receiver Meter Circuit - A d-c voltmeter meeting the requirements of MIL-M-6 shall be installed in the front panel. By the use of a meter receiver switch, paragraph 3.3.1.4.7.(21), located on the front panel, any of the following items may be monitored.

(1) Receiver Voltage - With the meter switch in this position, the panel meter shall indicate the supply voltage applied to the receiver amplifier terminals. (The meter shall be calibrated from 0-30 volts.) The voltage applied to this receiver may be varied from 22-30 volts by a potentiometer control on the front panel.

(2) Receiver Current - With the meter switch in this position, the panel meter shall measure the power supply current being supplied to the receiver-amplifier. (The meter scale shall be calibrated from 0-3 amps).

(3) Relay Current - With the meter switch in this position, the panel meter shall measure the current in the coils of the parachute and space channel relays of the receiver-amplifier. (The meter scale shall be calibrated from 0-30 milli-amps.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:26Z
Check the source to verify that this is the current version before use.

3.3.1.4.2 Pulse Length Indicator - A meter with appropriate scale markings, meeting the requirements of specification MIL-M-6, shall be installed in the panel to indicate the following functions by means of a rotary selector switch, paragraph 3.3.1.4.7(6), located on the front panel.

(1) Coder Output Level - The indicator shall provide a means of measuring the coder output voltage to provide the proper frequency deviation in the transmitter.

(2) Pulse Length - The pulse length shall be indicated as the percent duty cycle of the audio pulses from the audio coder. The necessary circuitry to transulate the pulse length into a d-c voltage for the meter shall be incorporated in the test set.

(3) Receiver Tuning - The indicator shall give an indication for tuning the RF and IF circuits of the receiver-amplifier.

3.3.1.4.3 Gyro Simulator - Two precision wire wound potentiometers shall be installed in the front panel to simulate the electrical action of the roll potentiometer of a type K-7, or similar gyro vertical, and to simulate the electrical action of a pitch potentiometer. Both potentiometers shall be connected to the "Gyro" connector on the panel in the same manner as the potentiometers of the gyro vertical.

3.3.1.4.4 Sensitivity Indicator - The sensitivity indicator shall consist of two 950 ohm ± 1% resistors and a potentiometer, paragraph 3.3.1.4.7(2), in the center of the circuit and mounted on the panel. This circuit shall permit checking the sensitivity of the receiver roll and pitch channels to slight changes of simulated gyro vertical roll and pitch correction commands.

3.3.1.4.5 Servo Simulator - Two precision wire wound potentiometers of 1200 ohms each shall be used to simulate the electrical action of the servo follow-up potentiometer of a type D-9 servo actuator. One dial shall be labeled "ROLL SERVO" and the other "PITCH SERVO". The associated potentiometers shall be connected to the appropriate connector on the front panel.

3.3.1.4.6 Micropositioner Test - Provisions shall be included for installing and testing for proper operation of Barber-Colman type AYLZ 3157S micropositioners or approved equivalents.

3.3.1.4.7 Controls - The equipment shall include the following front panel controls:

(1) Roll Gyro Potentiometer - A precision wire wound potentiometer of 2000 ohms fitted with a large calibrated dial, with appropriate markings to indicate the equivalent angle of roll represented by the wiper on the element, shall be provided. This potentiometer is to simulate the electrical action of the roll potentiometer of a type K-7 or an approved equivalent and shall be marked "ROLL GYRO".

(2) Sensitivity Indicator - A precision potentiometer of 100 ohms shall be provided. The potentiometer shall be fitted with a large calibrated dial with markings indicating the equivalent roll or pitch angle represented by the positions of the potentiometer wiper and shall be marked "SENSITIVE GYRO".

(3) Pitch Gyro Potentiometer - A precision wire wound potentiometer of 2000 ohms marked "PITCH GYRO" shall be provided. The potentiometer shall be provided with a large calibrated dial with markings indicating the equivalent angle of pitch represented by the position of the wiper.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:26Z
Check the source to verify that this is the current version before use.

(4) Roll Servo Potentiometer - A precision wire wound potentiometer of 2000 ohms marked "ROLL SERVO" shall be provided. The potentiometer shall be fitted with a large calibrated dial with markings to indicate the equivalent angle of servo movement represented by the wiper.

(5) Pitch Servo Potentiometer - A precision wire wound potentiometer of 1200 ohms marked "PITCH SERVO" shall be provided. The potentiometer shall be fitted with a large calibrated dial with markings to indicate the equivalent angle of servo movement represented by the position of the wiper.

(6) Pulse Length Meter Switch - An eight section ten position rotary switch marked as noted below, shall be provided to monitor the following various functions of the test set:

(a) "CODER LEVEL" - The audio frequency coder output level.

(b) "ROLL CH" - Roll channel pulse duty cycle.

(c) "PITCH CH" - Pitch channel pulse duty cycle.

(d) "SPARE CH" - Spare channel output level.

(e) "O ADJUST" - Permits adjustment of meter for zero scale reading.

(f) "100% ADJUST" - Permits adjustment of meter for full scale reading.

(g) "M. P. TEST" - Microposition test.

(h) "GYRO NULL" - The null of gyro vertical in roll and pitch

(i) "RECEIVER TUNE" - Signal level at limiter of Receiver R-602/ARW.

(j) "METER OFF" - Switch off meter.

(7) Meter "0 ADJUST" Control - A 1000 ohm potentiometer marked "0 ADJUST" shall be provided for adjustment of the pulse length meter for zero reading.

(8) Meter "100% ADJUST" Control - A 100,000 ohm potentiometer marked "100% ADJUST" shall be provided to permit adjustment of the pulse length meter for full scale deflection.

(9) Micropositioner Test Switch - A SPDT switch marked "MICROPOSITIONER TEST" shall be provided to control the current flowing through the micropositioner windings during test.

(10) Micropositioner Potentiometer Control - A potentiometer of 1000 ohms shall be provided to control the current flowing through the micropositioner windings during test.

(11) Micropositioner Indicator - Two indicator lights shall be provided to permit a visual check of the operation of the micropositioner switching contacts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:26Z
Check the source to verify that this is the current version before use.

(12) Gyro Power Switch - A SPST switch marked "GYRO ON" and "OFF" shall be provided for application of primary power to the motors of the gyro vertical.

(13) Gyro Indicators - Two indicator lamps marked "CHUTE" and "SPARE" shall be provided to permit a visual check on the operation of the receiver parachute and spare channel relays.

(14) Gyro Sensitivity Switch - A four pole three position switch marked "SENSITIVE ROLL", "NORMAL" and "SENSITIVE PITCH" shall be provided to connect the gyro sensitivity circuit in place of the gyro roll or gyro pitch simulator potentiometers.

(15) Auto Pilot-Gyro Switch - Two four pole DT switches, one marked "ROLL" and the other "PITCH" shall be provided to simulate either autopilot or gyro test operation. These switches shall independently switch the roll and pick channels.

(16) Servo Centering Switch - Two SPST switches marked "SERVO CENTERING" shall be provided to permit centering of roll and pitch servos respectively.

(17) Servo Simulator Switches - Two TPDT switches each marked "SERVO" - "SIMULATED" shall be provided to permit the use of D-9 servos in an actual circuit or simulate servo operation by using the servo simulator controls.

(18) Servo Relay Indicators - Two indicator lights shall be provided above each of the servo simulator potentiometers. They shall be connected to the servo connectors on the front panel to indicate closure of the polarized relay contacts and resultant direction of servo rotation.

(19) Power Switches - A SPST toggle switch marked "ON" and "OFF" shall be provided to control the application of power to the test set.

(20) Power Indicator - An indicator lamp marked "POWER" shall be provided to indicate when power is applied.

(21) Receiver Meter Switch - A three position switch marked "RELAY CURRENT 30 MA. SCALE", "RECEIVER VOLTAGE 30 VOLT SCALE" and "RECEIVER CURRENT 3A SCALE" shall be provided to permit receiver meter to monitor receiver input voltage current, and parachute and spare channel relay currents.

(22) Receiver Voltage Adjust - A 8 ohm potentiometer marked "lower-raise" shall be provided to permit variance of the receiver input voltage.

3.3.1.4.8 Connectors - The equipment shall provide the following front panel connectors:

(1) Receiver Power Connector - An AN-3102A-28-15S connector receptacle per MIL-C-5015 and marked "RECEIVER POWER" - "J-101" shall be provided for connection to the appropriate receptacles of the R-602/ARW receiver being tested.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:26Z
Check the source to verify that this is the current version before use.

(2) Coder Power Connector - An AN-3102A-18-1P connector receptacle per MIL-C-5015 and marked "CODER POWER" -"J-102" shall be provided for connection to the power receptacle J101 of the coder being tested.

(3) Power Input - An AN-3102A-16-11P connector receptacle per MIL-C-5015 marked "POWER INPUT 50 VOLTS D. C. " -"J-103" shall be provided for the connectors of the primary power to the tester.

(4) Coder Audio Input - A UG-657/U connector receptacle per MIL-C-3806 marked "CODER AUDIO INPUT" shall be provided for connection to audio connector of coder being tested.

(5) Signal Generator Modulating Input - A UG-657/U connector receptacle per MIL-C-3806 marked 'SIG. GEN MOD. INPUT" -"J105" shall be provided for connecting a modulating signal derived from coder receptacle of AN/ARM-19 Signal Generator.

(6) Receiver Audio Output - A JJ-034 telephone jack per MIL-P-642 marked "RECEIVER OUTPUT" shall be provided for aural monitoring of the command tones.

(7) Micropositioner Test Socket - An eight pin socket shall be provided for connection to the micropositioner being tested.

3.3.2 Cover, Test Set CW-(*)/ARM-28 - The test set cover shall be detachable and shall provide space for stowing the handbook and cable assemblies. The cover shall be provided with a hinged plate, over the cable compartment, which is secured with a thumb type lock fastener. A seal in the cover shall make the case watertight when the cover is clamped closed. The cover shall be fitted with four rubber feet for mounting on irregular surfaces.

3.3.3 Electrical Power Cable Assembly CX-(*)/U (8'-0") - The equipment shall be furnished with an eight foot length of two conductor cable CX-(*)/U terminated at one end with an AN-3102A-16-11P connector per MIL-C-5015 and the other end unterminated.

3.3.4 Cable, Special Purpose, Electrical CX-(*)/U(5'-0") - The equipment shall be furnished with a five foot length of seven conductor cable terminated at one end with an AN-3106A-18-1S connector per MIL-C-5015 and the other end with an AN-3108A-18-1S connector per MIL-C-5015, to supply power to KY-133/ARW-66 coder.

3.3.5 Cable, Special Power, Electrical CX-(*)/U(5'-0") - The equipment shall be furnished with a receiver power cable a five foot long terminated in an AN-3106-28-15P, per MIL-C-5015, plug on one end. The other end shall be a confluent cable assembly consisting of six cables as follows:

(1) Chute Cable - A two conductor cable terminating in an AN-3106A-10SL-4S plug per MIL-C-5015.

(2) Servo Cable - A two conductor cable terminating in a connector plug AN-3106A-12SL-3P per MIL-C-5015.

(3) Power Cable - A seven conductor cable terminating in a connector plug AN-3106A-16S-1S per MIL-C-5015.

(4) Gyro Cable - A seven conductor cable terminating in a connector plug AN-3106A-16S-1P per MIL-C-5015.

(5) Elevator Cable - A seven conductor cable terminating in a connector plug AN-3106A-16S-1P per MIL-C-5015.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:26Z
Check the source to verify that this is the current version before use.

(6) Aileron Cable - A seven conductor cable terminating in a connector plug AN-3106A-16S-1P per MIL-C-5015.

3.3.6 Servo Test Mount MT-(*)/U - A mounting bracket MT-(*)/U shall be provided for supporting two type D-9 servos. Appropriate scales and pointers shall be installed to indicate angular displacement of the servo power shaft from neutral. When mount is not being used, it is to be set in a cutout in the upper right hand portion of the front panel.

3.3.7 Gyro Test Mount MT-(*)/U - A mounting bracket MT-(*)/U shall be provided. This mount shall have attached to its underside bracketry and tilt table for mounting and checking operation of the gyro-vertical. When mount is not in use it is set in a cutout in the upper left hand portion of the front panel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:26Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests

(1) Preproduction Tests - Preproduction Tests are those tests conducted on an equipment representative of the equipments to be supplied under the contract, for the purpose of determining that the equipment meets all the requirements of the applicable specifications. Preproduction Tests shall be accomplished by Contractor Demonstration Tests and Service Approval Tests.

(a) Contractor Demonstration Tests - Contractor Demonstration Tests shall be accomplished under the responsibility of the contractor to demonstrate that the equipment complies with all applicable requirements.

(b) Service Approval Tests - Service Approval Tests shall be accomplished under the responsibility of the procuring service for the purpose of verifying the results of the Contractors Demonstration Tests and for conducting any additional tests deemed necessary to determine that the equipment complies with all applicable requirements.

(2) Acceptance Tests - Acceptance Tests are those tests accomplished on equipment submitted for acceptance under the contract. Acceptance Tests shall consist of the following:

(a) Individual Tests - Individual Tests are those tests conducted on each equipment submitted for acceptance under the contract.

(b) Sampling Tests - Sampling Tests are those tests accomplished on a quantity of equipments as approved by the Bureau of Aeronautics and which have previously passed Individual Tests.

(c) Special Tests - Special Tests are those tests accomplished on a quantity of equipments as approved by the Bureau of Aeronautics.

4.2 Preproduction Tests -

4.2.1 Contractor Demonstration Tests - The contractor shall conduct tests on one or more equipments as necessary to determine that the model as proposed by the contractor will meet all the requirements of the applicable specifications and the contract. Contractor Demonstration Tests shall be conducted in accordance with the approved test procedure. The data obtained by the contractor in conducting these tests shall be included with the design data submitted with the equipment for Service Approval Tests. The Inspector of Naval Material and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. When tests are supervised the Bureau of Aeronautics may, when conducting Service Approval Tests, omit all tests which will duplicate tests previously conducted by the contractor; however, in conducting Service Approval Tests, the Bureau of Aeronautics reserves the right to repeat any test previously conducted when deemed necessary. Contractors not having laboratory facilities to conduct all required tests shall obtain either the services of a commercial testing laboratory or written approval from the Bureau of Aeronautics to omit that portion of the tests which cannot be conducted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:26Z
Check the source to verify that this is the current version before use.

4.2.2 Service Approval Tests - The Service Approval Sample shall consist of one equipment representative of the production equipment to be supplied under the contract. This equipment shall be delivered by the contractor to the destination specified by the Bureau of Aeronautics.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for Service Approval Tests, the contractor shall also submit the accessory material and design and test data specified in MIL-E-5400. The design and test data shall indicate the physical and electrical characteristics of the equipment and establish that the equipment meets all applicable requirements. The test data shall include data obtained by the contractor in testing to MIL-T-5422 and MIL-T-945.

4.2.3 Scope of Tests - Preproduction Tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction Tests shall include environmental tests in accordance with MIL-T-945 and interference tests in accordance with MIL-I-6181. Shock tests shall be in accordance with MIL-T-5422.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No additional equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of any equipments prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance under the contract. The preproduction models shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the required Acceptance Tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Bureau of Aeronautics shall engage the services of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all Acceptance Tests required by this specification and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Acceptance Tests shall consist of the following:

(1) Individual Tests - Individual Tests shall be conducted on each equipment submitted for acceptance under the contract.

(2) Sampling Tests - Sampling Tests shall be conducted on a quantity of equipments as specified in 4.3.2.1. Equipments for sampling tests shall be selected by the Government Inspector after the equipment has passed the Individual Tests and has been packaged and packed ready for shipment.

(3) Special Tests - Special Tests shall be conducted on a quantity of equipment as approved or requested by the Bureau of Aeronautics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:26Z
Check the source to verify that this is the current version before use.

4.3.1 Individual Test - Each equipment submitted for acceptance shall be subjected to the Individual Test to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum each equipment accepted must successfully pass the tests specified in the following paragraph.

4.3.1.1 Scope of Test - As a minimum, each equipment shall be subjected to the following:

(1) Examination of Product

(2) Operational Test

(3) Reliability Test

4.3.1.1.1 Examination of Product - Each equipment shall be carefully examined to determine that the requirements for material and workmanship as specified in MIL-E-5400 have been met. Also, it shall be determined that approval has been granted for the use of all non-standard parts.

4.3.1.1.2 Operational Test - Each equipment shall be operated. The period of operation shall be long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.1.3 Reliability Test - Each equipment shall be operated (without isolators) under the conditions specified below for a period of 12 hours without a failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only such adjustments shall be permitted which can be made using such controls and adjustments which would normally be accessible during flight.

Temperature	Ambient Room
Humidity	Ambient Room
Vibration (10 minutes)	*Any selected frequency within the range of 20 to 60 cps (excluding resonant points) and an amplitude of ± 3 g's*

The ten (10) minute vibration run shall be made at the beginning of the twelve (12) hour period. The equipment shall be nonoperating during the vibration run. During the twelve (12) hour period of operation the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur; it shall be repaired and the test started over, except the ten (10) minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action must be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The twelve (12) hour period as specified above may be composed of several short periods of three (3) hours or more.

4.3.2 Sampling Test - Sampling Tests shall be conducted on a quantity of equipment as specified in the following paragraph. Equipments selected must first have passed all of the individual tests. Sampling Tests shall be conducted on complete equipments.

4.3.2.1 Sampling Test Schedule - Sampling Tests shall be performed on the following quantity of equipment. Selection of equipment for tests shall be under the supervision of the naval inspector.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:26Z
Check the source to verify that this is the current version before use.

Quantity Delivered for Acceptance	Quantity to be tested
From the first 10	1
From 11 to 100	2
From 101 to 200	1
From 200 and above	1 for each additional 200 or fraction thereof

4.3.2.2. Scope of Test - As a minimum each equipment selected for Sampling Test shall be subject to the tests specified below.

Tests	
I	Conduct a complete operational test at ambient room conditions and nominal input voltages, making all necessary measurements to assure all applicable specification requirements have been met.
II	Repeat the above using the highest and lowest specified input voltages and frequencies.
III	Conduct the reliability test specified in paragraph 4.3.1.1.3 except the test duration shall be 50 hours and two failures shall be permitted.

4.3.3 Special Test - Special Tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. Equipment selected for Special Test may be selected from equipment previously subjected to the Sampling Test.

4.3.3.1 Special Test Schedule - In selecting equipment for Special Tests the following schedule shall be followed:

(1) An early production equipment.

(2) An equipment midway of the production run.

(3) An early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the Bureau of Aeronautics.

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory quality control.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:26Z
Check the source to verify that this is the current version before use.

4.3.3.2 Scope of Test - Special Tests shall consist of such tests as approved by the Bureau of Aeronautics. Test procedures previously approved for the Preproduction and Sampling Tests shall be used where applicable. When not applicable the contractor shall prepare a test procedure and submit to the Bureau of Aeronautics for approval prior to conducting the tests. As a minimum the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports only such tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (90 minutes cycling - 30 min. in each of 3 planes - Freq 5 to 500 cps ± 5g)- Equipment normally mounted

(3) Humidity Tests

4.3.4 Equipment Failure - Should a failure occur during either the Sampling or Special Test, the following action shall be taken.

(1) *Determine the cause of failure*

(2) Determine if the failure is an isolated case or design defect.

(3) Suggest corrective action for Bureau of Aeronautics' approval, designed to reduce the possibility of the same failure occuring in future tests.

(4) Where practical, include a test in the Individual Test to check all equipments for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Test Procedures - The procedures used for conducting Preproduction Tests, and Acceptance Tests shall be prepared by the contractor and sent to the Bureau of Aeronautics for approval. The right is reserved by the Bureau of Aeronautics or the Naval Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the Bureau of Aeronautics will furnish the procedures to be used for testing.

4.5 Pre-Submission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements. The design data submitted by the contractor for the preproduction model shall indicate its physical and electrical characteristics and other necessary test data which will establish that the equipment meets all applicable requirements.

4.6 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken o correct the defects found in the original shall be furnished the Inspector. Units rejected fter retest shall not be resubmitted without the specific approval of the procuring agency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:26Z
Check the source to verify that this is the current version before use.

4.7 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the Bureau of Aeronautics. To obtain Bureau of Aeronautics approval the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This applies for either a Bureau of Aeronautics' recommended change or a contractor's proposed change. When required the contractor shall also prepare an Electronic Material Change (EMC) and an Electronic Material Bulletin (EMB) to cover the approved change. ECP's, EMC's and EMB's shall be prepared in accordance with the requirements of SAR-300.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be packaged, packed and marked for shipment in accordance with the requirements of Specifications MIL-P-17555 and MIL-STD-129.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 30.0 - 0.5 V, DC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

PATENT NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:26Z
Check the source to verify that this is the current version before use.